Unit 6

Replacement Theory

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→ Replacement theory deals with determining the time at which a system, machine or equipment should be replaced, to minimize cost and maximize efficiency.

* Reasons for replacement

- Repair and maintenance costs increase with time
- New equipment is available that is cheaper, faster and more efficient.
- Sudden failures in items like bulbs, fuses, etc.

* Replacement Models:

1, Replacement of items whose maintenance cost increase with time / that deteriorate with time

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- Value of money doesn't change with time → don't replace
- Value of money change with time → replace when increasing (i.e. RC for next year > AC of selected year)

2, Replacement of items that fails suddenly

- Individual replacement policy
- Group replacement policy

* Replacement of items that deteriorate with time

→ In most of the cases maintenance costs of items increases with time and after a certain time a stage comes at which the maintenance cost is so large/high that it is more profitable to replace the old item.

- If the cost decrease or remain constant with time, the best decision is never to replace the item but this situation is rare. If these cost fluctuate with time, the item should be replaced only when they are increasing.

i.e. If maintenance cost for next year is more than the average annual cost of the selected year, then replace the machine/item at the end of the selected year.

- MC – maintenance cost
- CMC – Cumulative MC
- D – Depreciation (C−S)
- TC – Total cost (CMC + D)
- AC = Average cost (TC/n)
- n = no. of years

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Example

Q.1) The cost of a machine is RS. 61,000 and its scrap value is RS. 1000. The maintenance costs found from the past experience are as follows:

Year	1	2	3	4	5	6	7	8
Maintenance cost (RS.)	1000	2500	4000	6000	9000	12000	16000	20000

When should the machine be replaced?

Soln

Here, original cost (C) = RS. 61,000, scrap value (S) = RS. 1000

Years(n)	MC	CMC	Depreciation(D) = C−S	TC = CMS+D	AC = TC/n
1	1000	1000	61000 − 1000 = 6000	61000	61000
2	2500	3500	"	63500	31750
3	4000	7500	"	67500	22500
4	6000	13500	"	73500	18375
5	9000	22500	"	82500	16500
6	12000	34500	"	94500	15750
→ 7	16000	50500	"	110500	15785.71
8	20000	70500	"	130500	16312.5

From the table, we can observe that the average annual cost is minimum (RS. 15750) in the 6th year.

Hence, the machine should be replaced at the end of 6th year.

∵ Running cost for next year > Average AC for selected year

(i.e. 16000 > 15750)

Some related terms

- C = capital cost or original cost of an item
- S = Scrap value or Resale value / year-end trade in value
- n = no. of years the item would be in use
- MC/RC/OC = Maintenance cost / Running cost / operating cost
- AC = Average anual cost

⇒ Total cost = (capital cost − scrap value) + Maintenance cost

= Depreciation cost + M.C

⇒ Depreciation (D) = capital cost − scrap value

= C − S

⇒ MC/RC/OC = cost of spears + cost of labor

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Q.2. The original cost of a machine is Rs. 6200 and its scrap value is Rs. 200. The maintenance costs found from exprience are as follows:

Year	2014	2015	2016	2017	2018	2019	2020	2021
Maintenance cost (Rs.)	100	250	400	600	900	1200	1600	2000

When should the machine be replaced?

Soln

Given,

Original cost (C) = Rs. 6200

Scrap value (S) = Rs. 200

Years (n)	MC	CMC	Depreciation (D) = C−S	TC = CMC+D	AC = TC/n
1	100	100	6200 − 200 = 6000	6100	6100
2	250	350	"	6350	3175
3	400	750	"	6750	2250
4	600	1350	"	7350	1837.5
5	900	2250	"	8250	1650
6	1200	3450	"	9450	1575
7	1600	5050	"	11050	1578
8	2000	7050	"	13050	1635

From the table we can observe that average annual cost is minimum (Rs. 1575) on the sixth year. Hence, the machine should be replaced at the end of 6th year.

- OC - operating cost
- COC - cumulative OC
- S - Scrap / Resale value

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Q.3 A machine cost Rs. 10,000. Its operating cost and resale values are given below:

year	1	2	3	4	5	6	7	8
operating cost	1000	1200	1400	1700	2000	2500	3000	3500
Resale value	6000	4000	3200	2600	2500	2400	2000	1600

Determine at what time it should be replaced?

Soln

Given, original cost (C) = Rs. 10,000

Year(n)	OC	COC	Resale value(S)	D = C-S	TC = COC+D	AC = TC/n
1	1000	1000	6000	4000	5000	5000
2	1200	2200	4000	6000	8200	4100
3	1400	3600	3200	6800	10400	3466.7
4	1700	5300	2600	7400	12700	3175
5	2000	7300	2500	7500	14800	2960
6	2500	9800	2400	7600	17400	2900
7	3000	12800	2000	8000	20800	2971.4
8	3500	16300	1600	8400	24700	3087.5

From table, we can observe the average annual cost is minimum (Rs. 2900) at 6th year.

Hence, the machine should be replaced at the end of 6th year.

* Replacement of items that fails suddenly

- Individual replacement policy : under this policy an item may be replaced immediately after its failure.
- Group replacement policy : under this policy, the items are replaced in group after a certain period (say t) irrespective of whether they have failed or not. If any items fails before its preventive replacement is due (i.e. before the optimal time), it may be individually replaced.

We have optimal policy:

Group replacement must be made at the end of t^th period if the cost of individual replacement for the t^th period is greater than the avg. cost per period, through the end of 't' period.

Example:

Q.1, The following mortality rates have been observed for certain type of light bulbs.

Week	1	2	3	4	5.
percent failing by the end of week	10	25	50	80	100

- There are 1000 bulbs in use and its costs Rs. 2 to replace an individual bulb, which has burnt out. If all the bulbs were replaced simultaneously, it would cost 50 paisa per bulb. It is proposed to replace all bulbs at fixed intervals, whether or not they have brunt out and to continue replacing burnt out bulbs as they fail. At what intervals should all the bulbs be replaced?

Soln

Step-1: let p_i be the probability that a light bulb fails during the i^{th} week of its life.

$p_1 = 10/100 = 0.1$

$p_2 = (25-10)/100 = 0.15$

$p_3 = (50-25)/100 = 0.25$

$p_4 = (80-50)/100 = 0.3$

$p_5 = (100-80)/100 = 0.2$

since, the sum of all above probability is 1, all the probabilities higher than p_5 will be zero

Thus, all light bulbs are sure to burnout by the 5^{th} week.

Step-2: let N_i be the no. of replacement made at the end of i^{th} week when all 1000 bulbs are new initially. Then, we have

$N_0 = 1000$ (Initial no. of bulbs)

$N_1 = N_0 P_1 = 1000 \times 0.1 = 100$

$N_2 = N_0 P_2 + N_1 P_1 = 1000 \times 0.15 + 100 \times 0.1 = 160$

$N_3 = N_0 P_3 + N_1 P_2 + N_2 P_1 = 1000 \times 0.25 + 100 \times 0.15 + 160 \times 0.1 = 281$

$N_4 = N_0 P_4 + N_1 P_3 + N_2 P_2 + N_3 P_1 = 377$

$N_5 = N_0 P_5 + N_1 P_4 + N_2 P_3 + N_3 P_2 + N_4 P_1 = 350$

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Step-3: Calculate the expected life of each bulb

$$\text{Expected life} = \sum_{i=1}^{5} i P_i = 1\times P_1 + 2\times P_2 + 3\times P_3 + 4\times P_4 + 5\times P_5$$

$$= 1\times 0.1 + 2\times 0.15 + 3\times 0.25 + 4\times 0.3 + 5\times 0.2$$

$$= 3.35$$

$\therefore$ Avg. no. of failures per week $= \frac{1000}{3.35} = 299$ (approx)

$\therefore$ Cost of individual replacement $= 299 \times 2$

$= Rs.\ 598$

Step-4: Group replacement calculation

End of week (n)	Cost of individual replacement	Total cost (TC) (Individual + group)	Avg. cost per week $\left(\frac{TC}{n}\right)$
1	$100\times 2 = 200$	$200 + 1000\times 0.5 = 700$	$700/1 = 700$
2	$160\times 2 = 320$	$320 + 700 = 1020$	$1020/2 = 510$
3	$281\times 2 = 562$	$562 + 1020 = 1582$	$1582/3 = 527.3$
4	$377\times 2 = 754$	$754 + 1582 = 2336$	$2336/4 = 584$
5	$350\times 2 = 700$	$700 + 2336 = 3036$	$3036/5 = 607.2$

Since, the average cost is minimum in the 2nd week so, the optimal replacement period to have a group replacement is after every 2nd week.

$\Rightarrow$ Since, the avg. cost is less than individual replacement cost (Rs. 598), so the group replacement policy is preferable.

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Q.2, Find the cost per period of individual replacement policy of an installation of 3000 street lamps, given below:
a) cost of replacing an individual bulb is Rs. 5
b) Cumulative probability of failure is given below

Weeks	0	1	2	3	4	5
Cumulative prob. of failure	0	0.1	0.3	0.5	0.7	1.0

Soln:
Let, P_i be the probability that a lamp bulb fails during the i^{th} week of its life. Then,

$P_1 = 0.1$ $\quad$ $P_4 = 0.7 - 0.5 = 0.2$
$P_2 = 0.3 - 0.1 = 0.2$ $\quad$ $P_5 = 1.0 - 0.7 = 0.3$
$P_3 = 0.5 - 0.3 = 0.2$

Since, the sum of all above probabilities is 1. so, the probabilities greater than P_5 are zero
Thus, all lamp bulbs are sure to burn out by 5^{th} week.

Let, N_i be the no. of replacement made at the end of i^{th} week
Thus,
$N_0 = 3000$ (Initial no. of bulb)
$N_1 = N_0P_1 = 3000 \times 0.1 = 300$
$N_2 = N_0P_2 + N_1P_1 = 3000 \times 0.2 + 300 \times 0.1 = 630$
$N_3 = N_0P_3 + N_1P_2 + N_2P_1 = 3000 \times 0.2 + 300 \times 0.2 + 630 \times 0.1 = 723$
$N_4 = N_0P_4 + N_1P_3 + N_2P_2 + N_3P_1 = 858.3$
$N_5 = N_0P_5 + N_1P_4 + N_2P_3 + N_3P_2 + N_4P_1 = 1316.43$
Now,
Expected life of each bulb $= \sum_{i=1}^{5} iP_i$
$= 1 \times 0.1 + 2 \times 0.2 + 3 \times 0.2 + 4 \times 0.2 + 5 \times 0.3$
$= 3.4$

$\therefore$ Average no. of failure per week $= \frac{3000}{3.4} = 882$ (approx)

$\therefore$ cost of individual replacement $= 882 \times$ Rs. 5
= Rs. 4410 //

• Annual service charge → Maintenance cost (MC)

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Q.8 A scooter with 1st cost Rs. 80,000 has depreciation and service pattern as given below:

	Year	1	2	3	4	5	6
D	Depreciation during year	28000	20000	14000	5000	4000	4000
MC	Annual service charge	18000	21000	25000	29000	34000	40000

How many years should the scooter be kept in service before replacement.

Soln,

Given, original cost (C) = 80,000

Years(n)	MC	CMC	Depreciation(D)	TC = CMC+D	AC = TC/n
1	18000	18000	28,000	46,000	46,000
2	21,000	39,000	20,000	59,000	29,500
3	25,000	64,000	14,000	78,000	26,000
4	29000	93,000	5,000	98,000	24,500
5	34,000	1,27,000	4,000	131,000	26,200
6	40,000	1,67,000	4,000	1,71,000	28,500

From the table we can observe the avg. annual cost is min. at 4th year.

So, the scooter should be kept for 4 years before replacement.